HAMZA ELGHONEMY

CONTACT

Cairo

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SKILLS

Languages:
Python, C#, C++, HTML/CSS, JavaScript, TypeScript, SQL
Machine/Deep Learning Libraries:
Keras, Scikit-Learn, TensorFlow, PyTorch
DevOps & Infrastructure:
Docker, Kubernetes, Jenkins, Grafana, Prometheus, Harbor, Rancher
Technologies/Frameworks:
Git, GitHub, Linux, Express.js, .NET, Django, Streamlit

LICENSE AND CERTIFICATIONS

- Advanced Learning Algorithms, DeepLearning.AI/ Stanford, 2024-09-01, Artificial Neural Networks, Network Models, Deep Learning, Machine Learning
- Unsupervised Learning, Recommenders, Reinforcement Learning, DeepLearning.AI/ Stanford, 2024-09-01, Unsupervised Learning, Recommender Systems, K-means clustering
- Supervised Machine Learning: Regression and Classification, DeepLearning.AI/ Stanford, 2024-08-01, Machine Learning Regression Models, Applied Machine Learning Algorithms

COURSEWORK

- Data Structures
- Software Methodology
- Algorithms Analysis
- Database Management
- Artificial Intelligence
- Computer Architecture

Software Engineering student with practical experience in AI/ML, DevOps, and full-stack development, currently researching interpretable medical AI with the German Research Institute for AI. Skilled in building end-to-end solutions across infrastructure, data, and model development. Seeking opportunities to contribute to software engineering and AI-driven projects.

EXPERIENCE

June 2025 - July 2025
Software Engineering & DevOps Intern ***Deloitte Innovation Hub***, Cairo, Egypt

- Helped build an AI-driven financial analyser using Streamlit, with guardrails and prompt engineering for sector-specific insights.
- Supported CI/CD pipeline optimisation and deployed monitoring and container tools (Grafana, Prometheus, Harbour, Rancher, and Jenkins).
- Worked with Kubernetes by creating pods, namespaces, and applying containerisation and microservices best practices.

September 2024 - February 2025
AI Scientist Intern ***Egyptian Space Agency***, New Cairo, Egypt

- Prepared and optimized data preprocessing pipelines, including image normalization, augmentation, and strategic dataset splitting, which were critical in achieving high model performance.
- Developed and implemented deep learning models (CNNs and transfer learning with VGG16, DenseNet121, MobileNet, and MobileNetV2) for automated classification of solar radio burst spectrograms, achieving up to 86% accuracy on processed datasets.
- Applied advanced ensemble techniques (soft voting, stacking) using Keras and scikit-learn to significantly enhance classification accuracy and model robustness.

PROJECTS

AstroNews – NLP-Powered Space News Analyzer
Python, Transformers, PyTorch, Scikit-Learn | August 2025

- Built an NLP application for analyzing space-related news using state-of-the-art embedding models (BAAI/bge-small-en-v1.5) to extract deep semantic meaning from text.
- Implemented **semantic search** enabling users to retrieve articles based on contextual relevance rather than keywords, leveraging transformer-based embeddings.
- Developed **automated clustering and summarization pipelines** using scikit-learn and Transformers to group similar articles and generate concise summaries.

LinkUp – Full-Stack Social Networking Platform
TypeScript, Node.js, Express, MongoDB, Socket.IO | April 2025

- Engineered a real-time WebSocket system with Socket.IO powering private messaging, typing indicators, read receipts, and online presence tracking.

- Image Processing
- Signal Processing
- Computer Vision
- Application Development
- Healthcare Information Systems

WEBSITES, PORTFOLIOS AND PROFILES

- linkedin.com/in/hamzaelghonemy
- github.com/Hamza-Elghonemy

- Implemented secure, multi-method authentication using JWT tokens and Google OAuth to support flexible and protected user access.
- Optimized backend performance, including connection-state recovery and efficient data structures, enabling the platform to handle **thousands of concurrent users** with low latency.

EEG Eye State Classification

Python, Deep Learning, Signal Processing | March 2025

- Achieved **93% classification accuracy** in detecting open vs. closed eye states using a bidirectional LSTM deep learning model.
- Processed raw EEG signals with the MNE library, applying techniques such as bandpass filtering (0.5–30 Hz) to isolate key frequency bands.
- Performed statistical analysis to determine the most influential EEG channels, guiding feature selection and model refinement.

Thermal Imaging Predictive Modeling

Python, Scikit-Learn | March 2025

- Developed and compared multiple ML models (Random Forest, SVR, Decision Trees, Gradient Boosting) to analyze facial thermal patterns.
- Conducted extensive hyperparameter tuning, identifying **Gradient Boosting Regressor** as the top-performing model ($R^2 \approx 0.75$).
- Engineered features from raw thermal sensor data to predict temperature distributions across facial regions (e.g., inner canthus, forehead).
- Benchmark-tested models using datasets from different camera systems (FLIR vs. ICI), producing confusion matrices and classification reports to assess sensor reliability.

EDUCATION

06/2026

Bachelor of Science Software Engineering

Cairo University, Giza, Cairo

- Graduation Project: Predicting Glioma Progression Using Temporal MRI Data and Explainable AI (XAI)
- Designing novel white-box architecture integrating capsule networks and attention mechanisms for clinically faithful explanations.
- Training and validating models on LUMIERE dataset using AUC, sensitivity, and specificity metrics.
- Developing clinician-facing prototype with interactive visual explanations of tumor progression predictions.
- GPA: 3.47/4.0